# Abstract


- **Generative AI–based video creation system** that converts **natural language text prompts into realistic video sequences**.
- Utilizes **multimodal deep learning architectures** combined with **cloud computing** for scalable processing.
- Employs **generative video models** to produce **coherent and visually consistent outputs**.
- **Automates video production workflows**, minimizing or eliminating the need for human intervention.
- Supports **text-to-video generation enhanced with depth-based structural guidance**.
- Built on a **pre-trained Stable Video Diffusion (SVD) model** with **additional temporal modules** for motion continuity.
- Ensures **high-quality, temporally coherent, and consistent video outputs** suitable for automated content creation.

# Introduction

- Video content creation is traditionally a **time-consuming and labor-intensive process** that requires skilled professionals and expensive software tools.
- The increasing demand for digital media has created a need for **automated and scalable video generation solutions**.
- Recent advancements in **Generative Artificial Intelligence and diffusion models** have enabled the synthesis of visual content from text prompts.
- **Text-to-video generation** reduces manual effort but still faces challenges such as temporal inconsistency and limited control over motion.
- This project proposes an **automated text-to-image-to-video generation system** using **Stable Video Diffusion (SVD)**.
- The proposed system produces **high-quality, realistic, and watermark-free videos** through a cost-effective and cloud-based inference pipeline.

# Literature Survey

| S.No | Author & Year | Proposed Work / Objective | Technology / Model Used | Dataset Used | Performance Achieved | Limitations Identified |
|---|---|---|---|---|---|---|
| 1 | Ho et al. (2025) | High-definition video generation from text prompts | Diffusion Models (Imagen Video) | Large-scale internal video datasets | High visual quality HD videos | Extremely high computational cost, not publicly accessible |
| 2 | Singer et al. (2024) | Text-to-video generation without paired text-video data | Make-A-Video (Diffusion-based) | Image–text datasets | Realistic short videos | Requires large training resources, limited motion control |
| 3 | Esser et al. (2023) | Structure and content-guided video synthesis | Gen-1 (Diffusion + Control) | Mixed image-video datasets | Improved motion consistency | Needs structural guidance, complex pipeline |
| 4 | Xing et al. (2023) | Video generation using text and depth-based guidance | Make-Your-Video (Latent Diffusion) | Depth maps + image datasets | Better temporal coherence | Additional depth input required |
| 5 | Stability AI (2023) | Image-to-video generation with smooth temporal transitions | Stable Video Diffusion (SVD) | Image-based diffusion training | High-quality, coherent videos | Limited video length, image-based input required |
| 6 | Open-Sora Team (2024) | Open-source large-scale video generation framework | Transformer + Diffusion | Large video datasets | Long-form video generation | Very high GPU and memory requirements |

# Problem Identification

- Manual video creation is **time-consuming, costly, and requires skilled professionals**, making it unsuitable for scalable content generation.
- Existing text-to-video systems rely only on **text prompts**, which fail to precisely control motion, depth, and scene structure.
- Many generative video models suffer from **poor temporal coherence**, causing flickering and inconsistent frame-to-frame transitions.
- Training or fine-tuning video generation models requires **large datasets and high computational resources**, limiting accessibility for academic use.
- Current systems provide **limited automation and customization**, making real-time or on-demand video generation difficult.
- There is a lack of a **cost-effective, fully automated, and watermark-free solution** that can generate realistic and consistent videos.

# Objectives

- To design and implement an **automated text-to-image-to-video generation system** using generative AI techniques.
- To utilize **Stable Video Diffusion** for generating temporally coherent and visually realistic video sequences.
- To apply **NLP-based preprocessing** for extracting key information such as objects, style, and scene details from text prompts.
- To ensure **smooth frame-to-frame transitions** through temporal motion modeling in video generation.
- To develop a **cloud-based and scalable pipeline** for efficient video generation without manual intervention.
- To produce **high-quality, consistent, and watermark-free videos** suitable for academic and real-world applications.

# Methodology

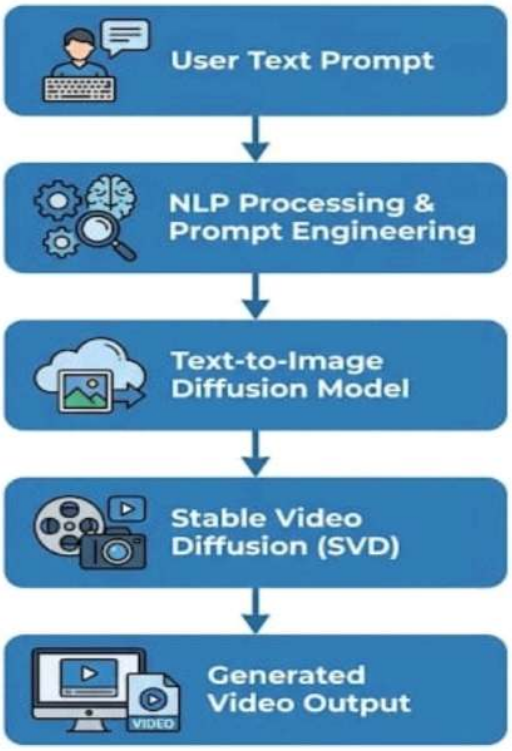
- **Text Input Acquisition**
  The system accepts a natural language text prompt from the user describing the desired scene, objects, motion, and visual style.
- **NLP-Based Prompt Processing**
  The input text is preprocessed using NLP techniques to extract meaningful keywords, scene attributes, and style information, ensuring accurate visual representation.
- **Text-to-Image Generation**
  The processed prompt is passed to a latent diffusion–based text-to-image model to generate a high-quality reference image that represents the initial video frame.
- **Image-to-Video Generation Using Stable Video Diffusion Model**
  The generated image is provided as input to the Stable Video Diffusion model, which applies temporal motion modeling to generate multiple consecutive frames.
- **Temporal Consistency Modeling**
  Stable Video Diffusion ensures smooth frame-to-frame transitions by learning temporal dependencies, resulting in coherent and realistic video motion.
- **Video Rendering and Storage**
  The generated frames are combined into a final video sequence and stored in cloud or local storage for playback and download.

# Methodology

**DATASETS:**

- **MSR-VTT (Kaggle)** provides ~10,000 short videos with high-quality human-written captions, suitable for **fine-tuning and evaluation**.
- **HowTo100M** contains large-scale instructional videos with ASR transcripts, from which a **small educational subset** is filtered.
- Videos are trimmed to **4–6 seconds**, resized to **256×256**, and paired with cleaned text prompts.
- Both datasets are converted into a **Stable Video Diffusion (SVD) compatible format**.
- Training is done using **LoRA fine-tuning** for memory-efficient learning on limited GPUs.
- This combination balances **caption quality (MSR-VTT)** and **educational diversity (HowTo100M)**.
- **MSR-VTT (**Microsoft Research Video**)** – from Kaggle: https://www.kaggle.com/datasets/vishnutheepb/msrvtt

# Block Diagram



# Materials and Methods

- **Programming Language:** Python, HTML, CSS, JavaScript
- **Deep Learning Framework:** PyTorch / TensorFlow
- **Hardware:** Cloud GPU or local GPU (AWS / NVIDIA GPU)
- **Generative Models:** Latent Diffusion Model (Text-to-Image), Stable Video Diffusion (Image-to-Video)
- **Backend Framework:** FastAPI / Flask
- **Storage:** Local storage or cloud storage
- **Libraries:** NumPy, OpenCV, Diffusers, Transformers

# User Interface

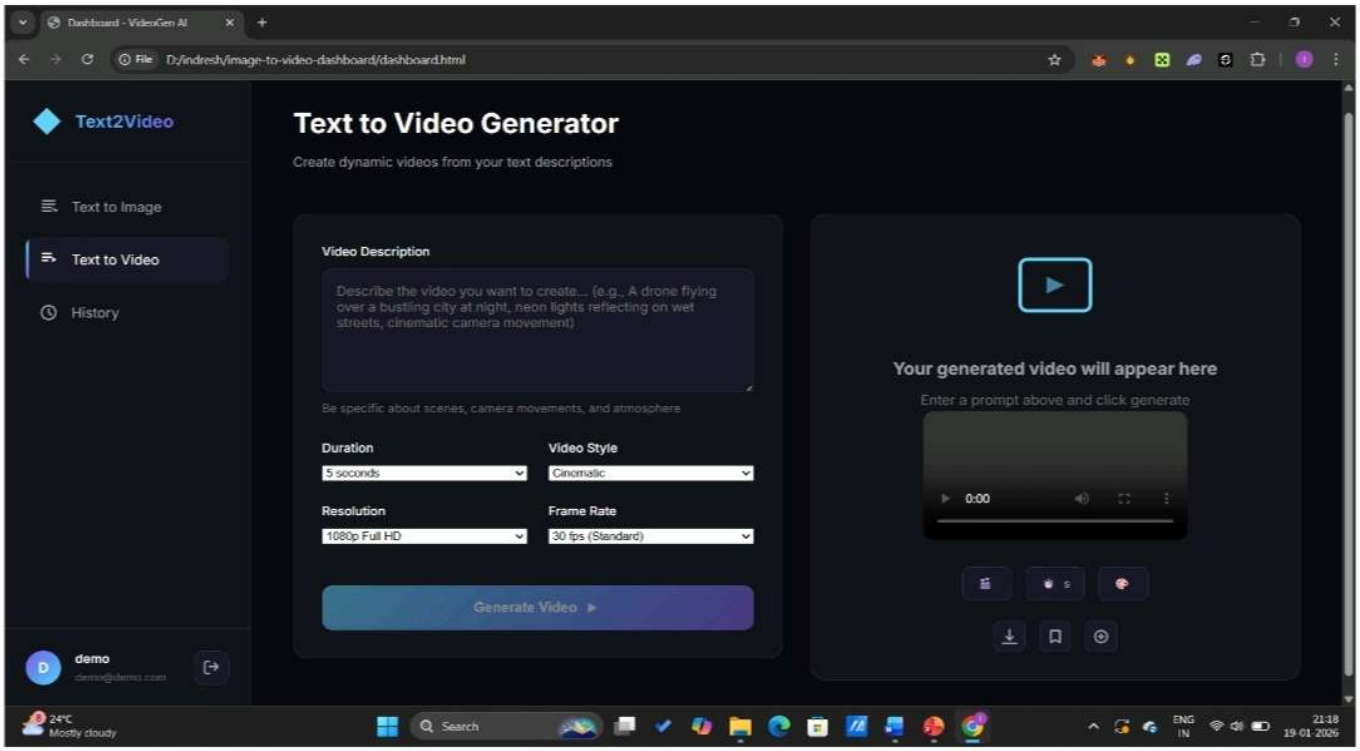



Figure1: Text To Video Generator – User Interface

# User Interface

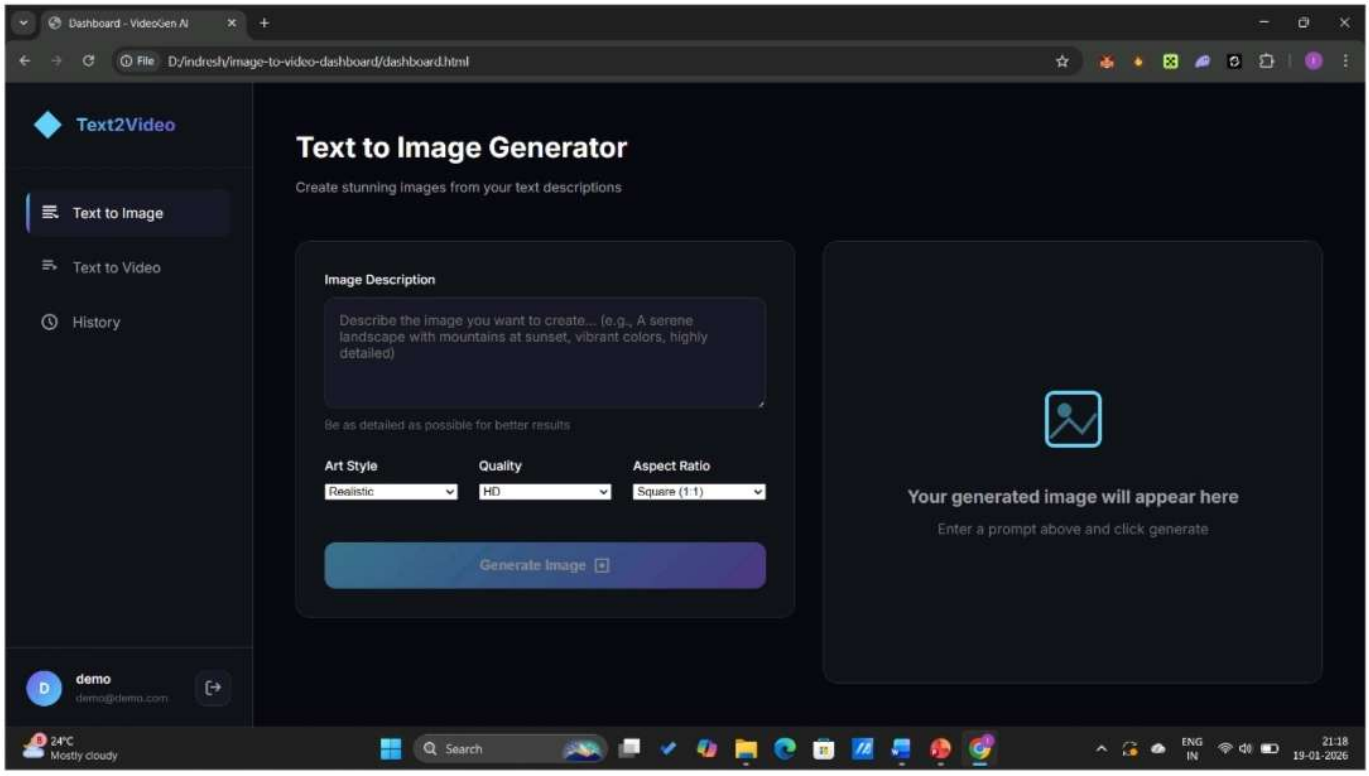



Figure2: Text To Image Generator – User Interface

# Results and Discussion

- The proposed system successfully generated realistic videos from text prompts using a text-to-image-to-video pipeline.
- Stable Video Diffusion produced **smooth and temporally coherent video sequences** with minimal flickering.
- The generated videos maintained **visual consistency** in objects, lighting, and scene structure across frames.
- The system operated efficiently in a cloud-based environment without requiring local model training.
- All outputs were **watermark-free**, making them suitable for academic and creative use.
- The results demonstrate the effectiveness of diffusion-based models for automated video generation.

# Conclusion

- This project successfully developed an **automated system** for generating videos from **text prompts** using **generative AI techniques**.
- The integration of **text-to-image diffusion** and **Stable Video Diffusion (SVD)** enabled the creation of **realistic** and **temporally coherent videos**.
- The proposed system significantly reduces **manual effort** and **processing time** compared to **traditional video production methods**.
- **Cloud-based execution** ensured **scalability** and **cost-effectiveness** without requiring **local model training**.
- The system generated **high-quality**, **consistent**, and **watermark-free video outputs**.
- Overall, the project demonstrates the **practical applicability** of **diffusion-based models** in **automated video content creation**.

# References


- Ho, J., et al.
  **"Imagen Video: High Definition Video Generation with Diffusion Models."**
  arXiv:2210.02303, 2025.
  (Referenced in related work)
- Singer, U., et al.
  **"Make-A-Video: Text-to-Video Generation Without Text-Video Data."**
  ICLR, 2023.
- Zhang, L., & Agrawala, M.
  **"ControlNet: Adding Conditional Control to Text-to-Image Diffusion Models."**
  arXiv:2302.05543, 2023.
- Esser, P., et al.
  **"Structure and Content-Guided Video Synthesis with Diffusion Models (Gen-1)."**
  arXiv:2302.03011, 2023.
- Stability AI, *"Stable Video Diffusion,"* https://stability.ai, 2023.
- Hugging Face, *"Diffusers Library Documentation,"* https://huggingface.co/docs/diffusers, 2023.